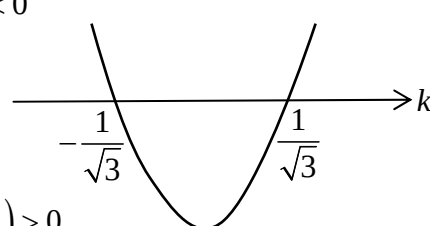
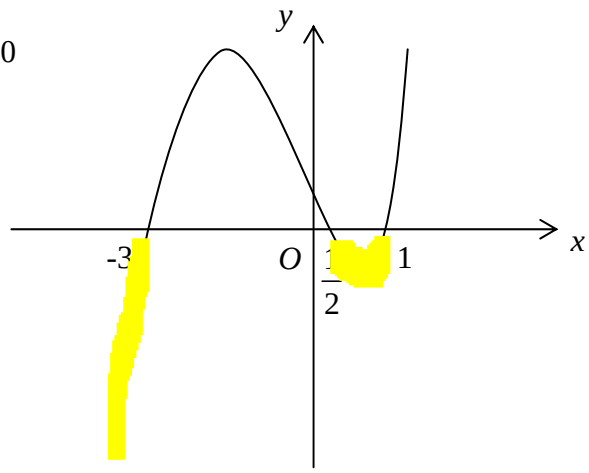


**Solutions to H1 Mathematics (8864) J2 Preliminary Examination 2013**

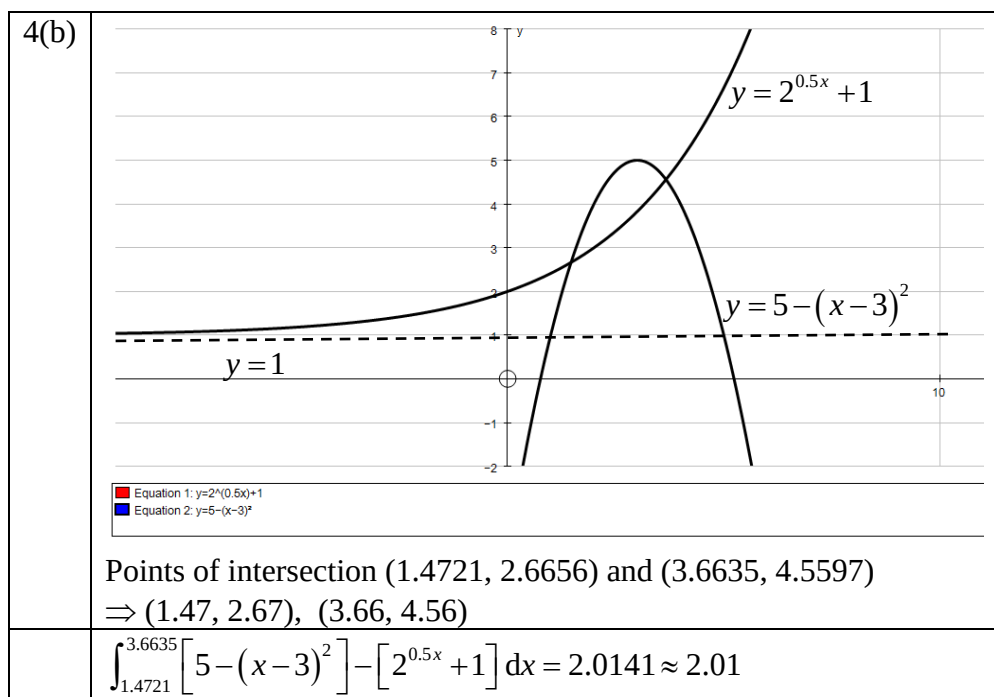
|      |                                                                                                                                                                                                                                                                                                                                                             |
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| 1(a) | $kx^2 - 2x + 3k < 0$ $k < 0 \quad \text{and} \quad (-2)^2 - 4(k)(3k) < 0$ $4 - 12k^2 < 0$ $1 - 3k^2 < 0$ $3k^2 - 1 > 0$ $(\sqrt{3}k - 1)(\sqrt{3}k + 1) > 0$ $k < -\frac{1}{\sqrt{3}} \quad \text{or} \quad k > \frac{1}{\sqrt{3}}$  $\therefore k < -\frac{1}{\sqrt{3}}$ |
| 1(b) | $e^{3x} - 1 = \frac{2}{e^{3x}}$ <p>Let <math>y = e^{3x}</math></p> $y - 1 = \frac{2}{y}$ $y^2 - y - 2 = 0$ $(y - 2)(y + 1) = 0$ $y = 2 \quad \text{or} \quad y = -1$ $e^{3x} = 2 \quad \text{or} \quad e^{3x} = -1 \text{ (reject)}$ $3x = \ln 2$ $x = \frac{1}{3} \ln 2$                                                                                   |
| 2.   | $-2x^3 - 3x^2 > 3 - 8x$ $2x^3 + 3x^2 - 8x + 3 < 0$                                                                                                                                                                                                                      |

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|  | $x < -3 \quad \text{or} \quad \frac{1}{2} < x < 1$ <p>Hence, solve <math>2(\ln x)^3 + 3(\ln x)^2 - 8\ln x + 3 &lt; 0</math></p> $\Rightarrow -2(\ln x)^3 - 3(\ln x)^2 > 3 - 8\ln x$ <p>Replace <math>x</math> by <math>\ln x</math>,</p> $\ln x < -3 \quad \text{or} \quad \frac{1}{2} < \ln x < 1$ $0 < x < e^{-3} \quad \text{or} \quad e^{\frac{1}{2}} < x < e$ |
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| 3. | <p>Area:</p> $\frac{1}{2}\pi x^2 + 2xy = 10$ $2xy = 10 - \frac{1}{2}\pi x^2$ $y = \frac{5}{x} - \frac{1}{4}\pi x$ <p>Cost <math>P = 15[2x + 2y] + 20\pi x</math></p> $= 30x + 30\left[\frac{5}{x} - \frac{1}{4}\pi x\right] + 20\pi x$ $= \frac{150}{x} + \left(30 + \frac{25}{2}\pi\right)x$ $= \frac{150}{x} + \left(\frac{60 + 25\pi}{2}\right)x$ <p>For cost to be minimum, <math>\frac{dP}{dx} = 0</math></p> $\frac{dP}{dx} = -\frac{150}{x^2} + \frac{60 + 25\pi}{2} = 0$ $\frac{150}{x^2} = \frac{60 + 25\pi}{2}$ $x^2 = \frac{300}{60 + 25\pi}$ $x^2 = \frac{60}{12 + 5\pi}$ $x = \sqrt{\frac{60}{12 + 5\pi}}, -\sqrt{\frac{60}{12 + 5\pi}} \text{ (N.A.)}$ |
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|  | $\frac{d^2P}{dx^2} = \frac{300}{x^3} > 0 \text{ when } x = \sqrt{\frac{60}{12+5\pi}}$ $\therefore \text{Cost is minimum when } x = \sqrt{\frac{60}{12+5\pi}}.$ $\min P = \frac{150}{\sqrt{\frac{60}{12+5\pi}}} + \left(\frac{60+25\pi}{2}\right)\sqrt{\frac{60}{12+5\pi}} = \$204 \text{ (nearest dollar)}$ |
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| 4(a) | $f(x) = \frac{1}{\sqrt{(1-2x)^3}}$ $\int f(x)dx = \int \frac{1}{\sqrt{(1-2x)^3}} dx = \int (1-2x)^{-\frac{3}{2}} dx = \frac{(1-2x)^{-\frac{1}{2}}}{\left(-\frac{1}{2}\right)(-2)} + C = \frac{1}{\sqrt{1-2x}}$ $\int_0^c \frac{1}{\sqrt{(1-2x)^3}} dx = 1.5$ $\left[ \frac{1}{\sqrt{1-2x}} \right]_0^c = 1.5$ $\frac{1}{\sqrt{1-2c}} - 1 = 1.5$ $\frac{1}{\sqrt{1-2c}} = 2.5$ $\frac{1}{1-2c} = \frac{25}{4}$ $1-2c = \frac{4}{25}$ $c = \frac{21}{50} \text{ (or 0.42)}$ |
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| 5(i)  | $y = 4 - \frac{a}{2x-1} = 4 - a(2x-1)^{-1}$ $\frac{dy}{dx} = 2a(2x-1)^{-2} = \frac{2a}{(2x-1)^2}$ <p>At P, <math>x = 3</math>, <math>\frac{dy}{dx} = \frac{2a}{(2(3)-1)^2} = \frac{2}{5}</math></p> $\frac{2a}{25} = \frac{2}{5}$ $a = 5 \quad [\text{AG}]$ |
| (ii)  | <p>At P, <math>x = 3, y = 3</math>, gradient of normal is <math>-\frac{5}{2}</math></p> $y - 3 = -\frac{5}{2}(x - 3)$ $y = -\frac{5}{2}x + 10\frac{1}{2}$                                                                                                   |
| (iii) | <p>When <math>y = 0</math>,</p> $0 = 4 - \frac{5}{2x-1}$ $2x - 1 = \frac{5}{4}$                                                                                                                                                                             |

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|       | $x = \frac{9}{8}$<br>$Q\left(\frac{9}{8}, 0\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| (iii) | $y = -\frac{5}{2}x + 10\frac{1}{2}$<br><br>When $x = 0, y = 10\frac{1}{2}$<br><br>Required Area = Area of trapezium $- \int_{\frac{9}{8}}^3 4 - \frac{5}{2x-1} dx$<br>$= \left( \frac{1}{2} \left( 10\frac{1}{2} + 3 \right) \times 3 \right) - \left[ 4x - \frac{5}{2} \ln(2x-1) \right]_{\frac{9}{8}}^3$ $= \frac{81}{4} - \left[ \left( 12 - \frac{5}{2} \ln 5 \right) - \left( \frac{9}{2} - \frac{5}{2} \ln \left( \frac{5}{4} \right) \right) \right]$ $= \frac{51}{4} - \frac{5}{2} (-\ln 5 + \ln 5 - \ln 4)$ $= \frac{51}{4} + \frac{5}{2} \ln 4$                                                                       |
| (iii) | <u>Alternative:</u><br><br>$y = -\frac{5}{2}x + 10\frac{1}{2}$<br><br>When $x = 0, y = 10\frac{1}{2}$<br><br>Required Area<br>$= \int_0^3 -\frac{5}{2}x + 10\frac{1}{2} dx - \int_{\frac{9}{8}}^3 4 - \frac{5}{2x-1} dx$ $= \left[ -\frac{5}{4}x^2 + 10\frac{1}{2}x \right]_0^3 - \left[ 4x - \frac{5}{2} \ln(2x-1) \right]_{\frac{9}{8}}^3$ $= \left[ -\frac{5}{4}(3)^2 + 10.5(3) \right] - [0] - \left[ \left( 12 - \frac{5}{2} \ln 5 \right) - \left( \frac{9}{2} - \frac{5}{2} \ln \left( \frac{5}{4} \right) \right) \right]$ $= \frac{51}{4} - \frac{5}{2} (-\ln 5 + \ln 5 - \ln 4)$ $= \frac{51}{4} + \frac{5}{2} \ln 4$ |

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| 6(i) | Select every $10^{\text{th}}$ (10% means every one out of 10) |
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|      | <p>or <math>k = \frac{\text{population size}}{\text{sample size}} = \frac{N}{0.1N} = 10</math>) electrical component from a production line, starting from say the 5<sup>th</sup> component, where the number 5 is randomly determined from the first 10 components produced. i.e choose the 5<sup>th</sup>, 15<sup>th</sup>, 25<sup>th</sup>,... until 10% of the components are selected.</p> |
| (ii) | <p>Choosing the sample using systematic sampling ensures that the sample chosen is evenly spread out and consists of components from different batches compared to choosing components from the first batch of sample.</p>                                                                                                                                                                      |

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| 7 | (i)   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|   | (ii)  | <p> <math>P(\text{Samuel wins a prize})</math><br/> <math>= P(B) + P(RB) + P(RRB)</math><br/> <math>= \frac{1}{25} + \left(\frac{24}{25}\right)\left(\frac{1}{24}\right) + \left(\frac{24}{25}\right)\left(\frac{23}{24}\right)\left(\frac{1}{23}\right)</math><br/> <math>= \frac{3}{25} \text{ (or 0.12)}</math> </p>                                                                                                                                                                         |
|   | (iii) | <p> <math>P(\text{took more than 1 ball} \mid \text{Samuel won a prize})</math><br/> <math>= \frac{P(\text{took more than 1 ball AND won a prize})}{P(\text{won a prize})}</math><br/> <math>= \frac{P(RB \text{ or } RRB)}{\frac{3}{25}}</math><br/> <math>= \frac{\left(\frac{24}{25}\right)\left(\frac{1}{24}\right) + \left(\frac{24}{25}\right)\left(\frac{23}{24}\right)\left(\frac{1}{23}\right)}{\frac{3}{25}}</math><br/> <math>= \frac{2}{3} \text{ (or 0.667 to 3 s.f.)}</math> </p> |
|   |       | <p> <math>P(\text{only one of them wins a prize}) = 3\left(\frac{3}{25}\right)\left(\frac{22}{25}\right)^2 = 0.278784</math> </p>                                                                                                                                                                                                                                                                                                                                                               |

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| 8 | (i) | $P(A B) = \frac{2}{3}p \Rightarrow \frac{P(A \cap B)}{P(B)} = \frac{2}{3}p$ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $1 - P(A' \cap B') = P(A) + P(B) - P(A \cap B)$ $1 - \frac{8}{15}p = 2p + p - P(A \cap B)$ $P(A \cap B) = \frac{53}{15}p - 1$ $\frac{P(A \cap B)}{P(B)} = \frac{2}{3}p$ $\frac{\frac{53}{15}p - 1}{p} = \frac{2}{3}p$ $\frac{53}{15}p - 1 = \frac{2}{3}p^2$ $53p - 15 = 10p^2$ $10p^2 - 53p + 15 = 0$ |
|   |     | $10p^2 - 53p + 15 = 0$ $p = \frac{3}{10} \quad \text{or} \quad p = 5 \text{ (NA since } 0 \leq p \leq 1)$ $P(\text{exactly one of A and B occurs})$ $= P(A \cup B) - P(A \cap B)$ $= \left[ 1 - \left( \frac{8}{15} \right) \left( \frac{3}{10} \right) \right] - \left[ \left( \frac{53}{15} \right) \left( \frac{3}{10} \right) - 1 \right]$ $= \frac{39}{50}$                                                            |

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| 9(i) | <p>Let <math>X</math> be the no. of seeds that germinates (out of 12).</p> <p><math>X \sim B(12, 0.65)</math></p> <p><math>P(X \geq 7) = 1 - P(X \leq 6) = 1 - 0.212735 = 0.78726 = 0.787 \quad (\text{to 3 s.f.})</math></p> |
| (ii) | <p>Let <math>X</math> be the no. of seeds that germinates (out of 12).</p> <p><math>X \sim B(12, 0.65)</math></p>                                                                                                             |

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|      | $P(4 \leq X \leq 9)$ $= P(X \leq 9) - P(X \leq 3)$ $= 0.8471 - 0.005698$ $= 0.84310$ $\approx 0.843 \text{ (to 3 s.f.)}$                                                                                                                                                                                                                                                                                                                                         |
| (ii) | <p><u>Alternative:</u></p> $P(4 \leq X \leq 9)$ $= P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7) + P(X = 8) + P(X = 9)$ $= 0.84310$ $\approx 0.843 \text{ (to 3 s.f.)}$                                                                                                                                                                                                                                                                                              |
|      | <p>Let <math>Y</math> be the no. of packets of seeds with fewer than 7 seeds that germinates (out of 25 packets).</p> $Y \sim B(25, 0.212735)$ $P(Y \leq 3) = 0.18982 \approx 0.190 \text{ (to 3 s.f.)}$                                                                                                                                                                                                                                                         |
|      | <p>Let <math>W</math> be the no. of seeds that will germinate (out of 120 seeds).</p> $W \sim B(120, 0.65)$ <p>Since <math>n = 120</math> is large,<br/> <math>np = 120(0.65) = 78 &gt; 5</math>, <math>nq = 120(0.35) = 42 &gt; 5</math></p> $W \sim N(120(0.65), 120(0.65)(0.35)) \quad \text{approximately}$ $W \sim N(78, 27.3) \quad \text{approximately}$ $P(W > 90) \xrightarrow{\text{c.c.}} P(W > 90.5) = 0.0083699 = 0.00837 \quad \text{(to 3 s.f.)}$ |

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| 10 | (i) | <p>Let <math>X</math> = mass of jam in a jar</p> $X \sim N(m, 6.5^2)$ $P(X < 320) \leq \frac{1}{1000}$ $P\left(Z < \frac{320 - m}{6.5}\right) \leq \frac{1}{1000}$ |
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|  |       | $\frac{320 - m}{6.5} \leq -3.0902$ $320 - m \leq -20.0863$ $m \geq 340.0863 \approx 340.09$ <p>Least <math>m = 340.09</math> (Shown)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|  | (ii)  | <p>Let <math>Y</math> = mass of an empty jar<br/> <math>Y \sim N(160, 4^2)</math><br/> <math>X \sim N(340.09, 6.5^2)</math><br/> <math>X + Y \sim N(340.09 + 160, 6.5^2 + 4^2) = N(500.09, 58.25)</math><br/> <math>P(X + Y \geq 505) = 0.26001 \approx 0.260</math> (to 3 s.f.)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  | (iii) | <p>Let <math>J = X + Y</math>. So <math>J \sim N(500.09, 58.25)</math>.<br/> <math>\bar{J} \sim N\left(500.09, \frac{58.25}{15}\right)</math><br/> <math>P(\bar{J} &gt; 498) = 0.85556 \approx 0.856</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|  | (iv)  | <p>Let <math>C</math> = mass of an empty large carton.<br/> <math>C \sim N(50, 1^2)</math><br/>         Let <math>D</math> = mass of an empty small carton.<br/> <math>D \sim N(20, 1^2)</math></p> <p>Let <math>L</math> = mass of a large carton of jam,<br/>         ie. <math>L = C + J_1 + J_2 + \dots + J_{12}</math><br/> <math>L \sim N(50 + 12 \times 500.09, 1 + 12 \times 58.25)</math><br/> <math>= N(6051.08, 700)</math></p> <p>Let <math>S</math> = mass of a small carton of jam,<br/>         ie. <math>S = D + J_1 + J_2 + \dots + J_6</math><br/> <math>S \sim N(20 + 6 \times 500.09, 1 + 6 \times 58.25)</math><br/> <math>= N(3020.54, 350.5)</math></p> <p><math>L - 2S \sim N(6051.08 - (2 \times 3020.54), 700 + 2^2(350.5))</math><br/> <math>= N(10, 2102)</math></p> <p><math>P(-10 &lt; L - 2S &lt; 10) = 0.16866 \approx 0.169</math> (to 3 s.f.)</p> |

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| 11 (i) | <p>Unbiased estimate of the population mean</p> $= \bar{x} = \frac{\sum(x-12)}{n} + 12 = \frac{105}{50} + 12 = 14.1$ <p>Unbiased estimate of the population variance</p> $s^2 = \frac{1}{n-1} \left[ \sum(x-c)^2 - \frac{(\sum(x-c))^2}{n} \right] = \frac{1}{49} \left( 2250 - \frac{105^2}{50} \right) =$ $\frac{2029.5}{49} = 41.4 \text{ (3 s.f.)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| (ii)   | <p>Let X be the percentage of fat content in burgers</p> <p><math>H_0: \mu = 12</math></p> <p><math>H_1: \mu &gt; 12</math></p> <p>Since <math>n</math> is large, by Central Limit Theorem,</p> $\bar{X} \sim N\left(12, \frac{2029.5}{2450}\right) \text{ approximately.}$ <p>Test statistic <math>Z = \frac{\bar{X} - 12}{\sqrt{\frac{2029.5}{2450}}} \sim N(0,1) \text{ approximately}</math></p> <p><math>\alpha = 0.02</math></p> <p>From GC, <math>z = \frac{14.1 - 12}{\sqrt{\frac{2029.5}{2450}}} = 2.307 \text{ (5 sf)}</math></p> <p><math>p\text{-value} = P(Z &gt; 2.307) = 0.0105 \text{ (3 sf)}</math></p> <p><math>\therefore</math> Since <math>p\text{-value} = 0.0105 \text{ (3 sf)} &lt; 0.02</math>, we <b>reject <math>H_0</math></b> at 2% level of significance and conclude there is <b>sufficient evidence</b> to refute the company's claim.</p> |
| (iii)  | <p>2% level of significance means there is a probability of 0.02 of wrongly concluding that the mean fat content is more than 12 % (rejecting <math>H_0</math>) when in fact it is 12%.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| (iv)   | <p>It is not necessary to assume a the percentage fat content have a normal distribution for the test to be valid, since sample size is large, the Central Limit Theorem applies.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|        | <p><math>H_0: \mu = 12</math></p> <p><math>H_1: \mu &gt; 12</math></p> <p>Since <math>n</math> is large, by Central Limit Theorem,</p> $\bar{X} \sim N\left(12, \frac{6.5^2}{50}\right) \text{ approximately.}$ <p>Test statistic <math>Z = \frac{\bar{X} - 12}{\sqrt{\frac{6.5^2}{50}}} \sim N(0,1) \text{ approximately}</math></p> <p>Claim is untrue <math>\Rightarrow</math> reject <math>H_0</math></p> <p>Reject <math>H_0</math> at 5% level of significance</p>                                                                                                                                                                                                                                                                                                                                                                                                   |

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|  | <p>[using GC: invNorm(0.95) = 1.64485]</p> <p><math>z \geq 1.64485</math></p> $\frac{F - 12}{\sqrt{\frac{6.5^2}{50}}} \geq 1.64485$ <p><math>F \geq 13.5</math></p> <p>Least possible value of <math>F</math> is <u>13.5</u></p> |  |
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| 12(i) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| (ii)  | <p><math>r = -0.993</math></p> <p>The value of <math>r</math> is close to <math>-1</math>. There is strong negative linear correlation between the amount of catalyst and the time taken for the chemical reaction. This means that the time taken for the chemical reaction decreases as the amount of catalyst present increases.</p>                                                                                                                                                           |
| (iii) | <p><math>t = 119.4385 - 40.5813x</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| (iv)  | <p><math>t</math> on <math>x</math> line is used since <math>x</math> is controlled and <math>t</math> is random.</p> <p><math>60 = 119.4385 - 40.5813x</math></p> <p><math>x = 1.4647 \approx 1.46</math> grams</p> <p>The estimate is reliable because <math>t = 60</math> is within the data range hence this is an interpolation and <math>r</math> is close to <math>-1</math>, indicating a strong negative correlation between the amount of catalyst and time taken for the reaction.</p> |